

Theoretical Physics for Mathematicians

Home Work 1

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Newtonian Space-Time and the Galilei Group

Exercise 1 ($\star$ The Galilei group). For $a \in \mathbb{R}$, $\vec{b}, \vec{v} \in \mathbb{R}^3$ and $R \in SO(3)$ define the map

$$g_{(a, \vec{b}, \vec{v}, R)} : \mathbb{R} \times \mathbb{R}^3 \longrightarrow \mathbb{R} \times \mathbb{R}^3, \quad (t, \vec{x}) \longmapsto (t + a, R\vec{x} + \vec{v}t + \vec{b}).$$

- (a) Show that the composition of two such maps is again of this form, and derive the composition law

$$g_{(a_2, \vec{b}_2, \vec{v}_2, R_2)} \circ g_{(a_1, \vec{b}_1, \vec{v}_1, R_1)} = g_{(a_1 + a_2, R_2 \vec{b}_1 + a_1 \vec{v}_2 + \vec{b}_2, R_2 \vec{v}_1 + \vec{v}_2, R_2 R_1)}.$$

Deduce that the set $\mathcal{G}$ of all these maps is a group, compute the inverse of $g_{(a, \vec{b}, \vec{v}, R)}$, and determine $\dim \mathcal{G}$.

- (b) Let

$$N = \left\{ g_{(a, \vec{b}, 0, I)} \mid a \in \mathbb{R}, \vec{b} \in \mathbb{R}^3 \right\}$$

be the subgroup of space-time translations. Show that N is normal in $\mathcal{G}$ and that $\mathcal{G}/N \cong SO(3) \ltimes \mathbb{R}^3$. Conclude that $\mathcal{G} \cong \mathbb{R}^4 \ltimes (SO(3) \ltimes \mathbb{R}^3)$.

- (c) Show that the boosts

$$B = \left\{ g_{(0, 0, \vec{v}, I)} \mid \vec{v} \in \mathbb{R}^3 \right\}$$

form an abelian subgroup isomorphic to $(\mathbb{R}^3, +)$, but that B is not normal in $\mathcal{G}$.

- (d) Let $\vec{x} : I \rightarrow \mathbb{E}^3$ be a trajectory and let $\tilde{\vec{x}}$ be its image under $g_{(a, \vec{b}, \vec{v}, R)}$. Show that

$$\ddot{\vec{x}} \equiv 0 \quad \text{if and only if} \quad \ddot{\tilde{\vec{x}}} \equiv 0.$$

Hence the notion of an inertial system is well defined on the whole group $\mathcal{G}$.

ANSWER:

(a) To show that the Galilean transformations are closed under composition.
Take

$$\begin{aligned} g_{(a_1, \hat{b}_1, \hat{v}_1, R_1)}(t, \bar{x}) &= (t + a_1, R_1 \bar{x} + \hat{v}_1 t + \hat{b}_1), \\ g_{(a_2, \hat{b}_2, \hat{v}_2, R_2)}(t, \bar{x}) &= (t + a_2, R_2 \bar{x} + \hat{v}_2 t + \hat{b}_2). \end{aligned}$$

Then

$$\begin{aligned} g_{(a_2, \hat{b}_2, \hat{v}_2, R_2)} \circ g_{(a_1, \hat{b}_1, \hat{v}_1, R_1)}(t, \bar{x}) &= g_{(a_2, \hat{b}_2, \hat{v}_2, R_2)}(t + a_1, R_1 \bar{x} + \hat{v}_1 t + \hat{b}_1) \\ &= \left(t + a_1 + a_2, R_2[R_1 \bar{x} + \hat{v}_1 t + \hat{b}_1] + \hat{v}_2(t + a_1) + \hat{b}_2 \right) \\ &= \left(t + a_1 + a_2, (R_2 R_1) \bar{x} + (R_2 \hat{v}_1) t + R_2 \hat{b}_1 + \hat{v}_2 t + \hat{v}_2 a_1 + \hat{b}_2 \right) \\ &= \left(t + a_1 + a_2, (R_2 R_1) \bar{x} + (R_2 \hat{v}_1 + \hat{v}_2) t + (R_2 \hat{b}_1 + \hat{v}_2 a_1 + \hat{b}_2) \right) \\ &= g_{(a_1 + a_2, R_2 \hat{b}_1 + \hat{v}_2 a_1 + \hat{b}_2, R_2 \hat{v}_1 + \hat{v}_2, R_2 R_1)}. \quad (*) \end{aligned}$$

Now $R_1, R_2 \in SO(3) \Rightarrow R_2 R_1 \in SO(3)$. Also: $R_2 \bar{v}_1 + \bar{v}_2 \in \mathbb{R}^3$, $R_2 \bar{b}_1 + \bar{v}_2 a + \bar{b}_2 \in \mathbb{R}^3$.
Hence (*) is also a Galilean transformation.

The identity element of this group will leave $(t, \bar{x})$ fixed. Thus:

$$\begin{aligned} g_{(a, \bar{b}, \bar{v}, R)}(t, \bar{x}) &= (t + a, R \bar{x} + \bar{v} t + \bar{b}) \\ &= (t, \bar{x}). \end{aligned}$$

It is clear that: $a = 0$, $\bar{v} = \bar{0}$, $\bar{b} = \bar{0}$ and $R = I$.

Hence $g_{(0, \bar{0}, \bar{0}, I)}$ is the identity element for the group $\mathcal{G}$.

Now let's find the inverse transformation for $g_{(a, \bar{b}, \bar{v}, R)} \in G$. Such that:

$$g_{(c, \bar{d}, \bar{u}, T)} \circ g_{(a, \bar{b}, \bar{v}, R)} = g_{(0, \bar{0}, \bar{0}, I)}. \quad (1)$$

$$g_{(a, \bar{b}, \bar{v}, R)} \circ g_{(c, \bar{d}, \bar{u}, T)} = g_{(0, \bar{0}, \bar{0}, I)}. \quad (2)$$

From (*) above we know that

$$\begin{aligned} g_{(c, \bar{d}, \bar{u}, T)} \circ g_{(a, \bar{b}, \bar{v}, R)} &= g_{(a+c, T\bar{b}+\bar{u}a+\bar{d}, T\bar{v}+\bar{u}, TR)} \\ &= g_{(0, \bar{0}, \bar{0}, I)}. \end{aligned}$$

Clearly:

$$\begin{aligned} a + c = 0 &\Rightarrow c = -a, \\ TR = I &\Rightarrow T = R^{-1} \quad (\text{possible as } R \in SO(3)), \\ T\bar{v} + \bar{u} = \bar{0} &\Rightarrow \bar{u} = -T\bar{v} = -R^{-1}\bar{v}, \\ T\bar{b} + \bar{u}a + \bar{d} = \bar{0} &\Rightarrow R^{-1}\bar{b} - (R^{-1}\bar{v})a + \bar{d} = \bar{0} \\ &\Rightarrow \bar{d} = R^{-1}(\bar{v}a - \bar{b}). \end{aligned}$$

The same result applies if we take (2). Hence

$$g_{(a,\bar{b},\bar{v},R)}^{-1} = g_{(-a, R^{-1}(\bar{v}a - \bar{b}), -R^{-1}\bar{v}, R^{-1})}.$$

The above three properties: closure, inverse and identity show that $\mathcal{G}$, the set of Galilean transformations, is indeed a group.

The total number of dimensions for $\mathcal{G}$ is $10 = 1$ (time) + 3 (position) + 3 (velocity) + 3 (rotation matrix).

(b)

I didn't know what a "semidirect product" was - I understand it as follows:

A **semidirect product** is a group G built from subgroups N and H such that

$$N \trianglelefteq G, \quad N \cap H = \{e\}, \quad G = NH.$$

Unlike a direct product, N and H need not commute. One subgroup acts on the other by conjugation.

To demonstrate that: $N = \{g_{(a,\bar{b},\bar{0},I)} \mid a \in \mathbb{R}, \bar{b} \in \mathbb{R}^3\}$ is normal in $\mathcal{G}$ we need to prove that: given any $g \in \mathcal{G}$ and $n \in N$, that:

$$g \circ n \circ g^{-1} \in N$$

This is just subscript manipulation using equation (*) repeatedly. Take $n = g_{(a,\bar{b},\bar{0},I)}$ and $g = g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)}$. Then:

$$\begin{aligned} g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)} \circ n \circ g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)}^{-1} &= g_{(a_2,\bar{b}_2,\bar{v}_2,R_2)} \circ g_{(a,\bar{b},\bar{0},I)} \circ g_{(-a_2,R_2^{-1}(\bar{v}_2 a_2 - \bar{b}_2), -R_2^{-1}\bar{v}_2, R_2^{-1})} \\ &= g_{(a_2+a, R_2\bar{b} + a\bar{v}_2 + \bar{b}_2, \bar{v}_2, R_2 I)} \circ g_{(-a_2, R_2^{-1}(\bar{v}_2 a_2 - \bar{b}_2), -R_2^{-1}\bar{v}_2, R_2^{-1})} \\ &= g_{(a_2+a, R_2\bar{b} + a\bar{v}_2 + \bar{b}_2, \bar{v}_2, R_2 I)} \circ g_{(-a_2, R_2^{-1}(\bar{v}_2 a_2 - \bar{b}_2), -R_2^{-1}\bar{v}_2, R_2^{-1})} \\ &= g_{(a, R_2\bar{b} + \bar{v}_2 a, \bar{0}, I)} \end{aligned}$$

Which is again an element of N . Hence N is a normal subgroup of $\mathcal{G}$.

Clearly N has dimension 4 (1-time and 3-position vector) - which leaves $10-4=6$ dimensions to be accounted for in $\mathcal{G}/N$. The remaining 6 dimensions are accounted for by the velocity and the rotations - which are independent, so: $SO(3) \ltimes \mathbb{R}^3$ and hence: $\mathcal{G}/N \cong SO(3) \ltimes \mathbb{R}^3$. It would be a simple procedure to show that there is an isomorphism between $\kappa : (\mathcal{G}/N, SO(3) \ltimes \mathbb{R}^3) \rightarrow \mathcal{G}$.

(c)

We will show that $B = \{g_{(0,\bar{0},\bar{v},I)} \mid \bar{v} \in \mathbb{R}^3\}$ is an abelian group.

This is simply demonstrated as follows (and using (*)):

$$\begin{aligned} g_{(0,\bar{0},\bar{v}_1,I)} \circ g_{(0,\bar{0},\bar{v}_2,I)} &= g_{(0,\bar{0},\bar{v}_2,I)} \circ g_{(0,\bar{0},\bar{v}_1,I)} \\ &= g_{(0,\bar{0},\bar{v}_1+\bar{v}_2,I)} \in B \end{aligned} \quad (**)$$

This shows that composition is both commutative and closed on the set B . The inverse of $g_{(0,\bar{0},\bar{v},I)}$ is obviously: $g_{(0,\bar{0},-\bar{v},I)}$. Hence B is an abelian group.

We can clearly define an isomorphism between $(\mathbb{R}^3, +)$ and $(B, \circ)$ - which is just a mapping between velocities and boosts as follows:

$$\phi : \mathbb{R}^3 \rightarrow B \text{ where: } \phi(\bar{v}) = g_{(0,\bar{0},\bar{v},I)}$$

From which:

$$\phi(v_1 + v_2) = g_{(0,\bar{0},\bar{v}_1+\bar{v}_2,I)} = g_{(0,\bar{0},\bar{v}_1,I)} \circ g_{(0,\bar{0},\bar{v}_2,I)} = \phi(v_1) \circ \phi(v_2)$$

Lastly, we must show that $B \not\trianglelefteq \mathcal{G}$.

We will proceed as we did in (b). Take any $g = g_{(a,\bar{b},\bar{u},R)} \in \mathcal{G}$ and $b = g_{(0,\bar{0},\bar{v},I)} \in B$; then:

$$\begin{aligned} g \circ b \circ g^{-1} &= g_{(a,\bar{b},\bar{u},R)} \circ g_{(0,\bar{0},\bar{v},I)} \circ g_{(-a,R^{-1}(\bar{u}a)-\bar{b},-R^{-1}\bar{u},R^{-1})} \\ &= g_{(a,\bar{b},R\bar{v}+\bar{u},R)} \circ g_{(-a,R^{-1}(\bar{u}a)-\bar{b},-R^{-1}\bar{u},R^{-1})} \\ &= g_{(0,-aR\bar{v},R\bar{v},I)} \notin B \end{aligned}$$

So, B is NOT a normal subgroup of $\mathcal{G}$.

(d)

For the given transformation we have:

$$t' = t + a \quad (3)$$

$$\vec{x}' = R\vec{x} + \vec{v}t + \vec{b} \quad (4)$$

(3) $\Rightarrow \Delta t = \Delta t'$ as time difference between any two events in the two different frames. Hence:

$$\frac{d}{dt'} = \frac{d}{dt}.$$

If we then differentiate (4) w.r.t time we get:

$$\dot{\vec{x}}' = \dot{R}\vec{x} + R\dot{\vec{x}} + \vec{v}$$

Again:

$$\ddot{\vec{x}}' = \ddot{R}\vec{x} + \dot{R}\dot{\vec{x}} + \dot{R}\dot{\vec{x}} + R\ddot{\vec{x}}$$

For a Galilean transformation $\dot{R} = 0$, otherwise we are not transforming between inertial frames.

Hence:

$$\ddot{x}' = R\ddot{x} \tag{5}$$

If $\ddot{x}' = \vec{0} \Rightarrow \ddot{x} = \vec{0}$, since R is not singular. Also $\ddot{x} = \vec{0} \Rightarrow \ddot{x}' = 0$ follows directly from (5). This means that Galilean transformations are valid transformations between inertial reference frames.

Exercise 4 ($\star$ Conservation laws for systems of point masses). Consider N point masses m_j with trajectories $\vec{x}_j(t) \in \mathbb{E}^3$ obeying

$$m_j \ddot{\vec{x}}_j = \vec{k}_j = \sum_{i \neq j} \vec{k}_{ij} + \vec{K}_j, \quad j = 1, \dots, N,$$

where $\vec{k}_{ij}$ is the internal force exerted on j by i and $\vec{K}_j$ is external.

Assume Newton's third law $\vec{k}_{ij} = -\vec{k}_{ji}$. Write $\vec{p}_j = m_j \dot{\vec{x}}_j$, $\vec{P} = \sum_j \vec{p}_j$, $M = \sum_j m_j$, $\vec{X} = M^{-1} \sum_j m_j \vec{x}_j$ and

$$\vec{J} = \sum_j \vec{x}_j \wedge \vec{p}_j.$$

- (a) Show that $\dot{\vec{P}} = \sum_j \vec{K}_j$, and deduce that $\vec{P}$ is conserved for a closed system.
(b) Deduce the centre-of-mass theorem: for a closed system

$$\vec{X}(t) = \vec{X}_0 + \vec{V}t.$$

- (c) Show that

$$\dot{\vec{J}} = \sum_j \vec{x}_j \wedge \vec{K}_j$$

provided the internal forces are central, i.e.

$$\vec{k}_{ji} = \frac{\vec{x}_j - \vec{x}_i}{|\vec{x}_j - \vec{x}_i|} f_{ji}(|\vec{x}_j - \vec{x}_i|).$$

Where exactly does centrality enter? Give an example of forces obeying $\vec{k}_{ji} = -\vec{k}_{ij}$ for which $\vec{J}$ is not conserved.

- (d) (Two-body reduction.) Let $N = 2$ with central internal forces and no external force. Set $\vec{r} = \vec{x}_1 - \vec{x}_2$ and $\mu = \frac{m_1 m_2}{M}$. Show that

$$\mu \ddot{\vec{r}} = \frac{\vec{r}}{|\vec{r}|} f_{12}(|\vec{r}|),$$

and prove König's decompositions

$$T = \frac{1}{2} M |\dot{\vec{X}}|^2 + \frac{1}{2} \mu |\dot{\vec{r}}|^2,$$

$$\vec{J} = M \vec{X} \wedge \dot{\vec{X}} + \mu \vec{r} \wedge \dot{\vec{r}}.$$

ANSWER:

(a)

$$\begin{aligned}
\vec{P} &= \sum_j \vec{p}_j \\
\Rightarrow \dot{\vec{P}} &= \sum_j \dot{\vec{p}}_j \\
&= \sum_j m_j \ddot{\vec{x}}_j \quad (\text{as } m_j \text{ is constants}) \\
&= \sum_j \left\{ \left[\sum_{i \neq j} \vec{k}_{ji} \right] + \vec{K}_j \right\}.
\end{aligned} \tag{4.1}$$

Now if we take $\vec{k}_{ii} = 0$ ($\forall i$) then (4.1) becomes,

$$\dot{\vec{P}} = \sum_j \left\{ \sum_i \vec{k}_{ji} + \vec{K}_j \right\}.$$

Since: $\vec{k}_{ij} = -\vec{k}_{ji} \Rightarrow \sum_j \sum_i \vec{k}_{ij} = \vec{0}$.

$$\therefore \dot{\vec{P}} = \sum_j \vec{K}_j.$$

For a closed system each $\vec{K}_j = \vec{0}$

$$\begin{aligned}
&\Rightarrow \dot{\vec{P}} = \vec{0} \\
&\Rightarrow \vec{P} \text{ is constant.}
\end{aligned}$$

(b)

$$\begin{aligned}
M\vec{X} &= M \left\{ \frac{1}{M} \sum_j m_j \vec{x}_j \right\} \\
&= \sum_j m_j \vec{x}_j
\end{aligned}$$

Differentiating w.r.t. time:

$$\begin{aligned}
M\dot{\vec{X}} &= \sum_j m_j \dot{\vec{x}}_j \\
&= \vec{P} \quad (\text{a constant}). \\
&= M\vec{V} \quad (\text{where } \vec{V} \text{ is a constant}).
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \dot{\vec{X}} = \vec{V} \\
&\Rightarrow \vec{X} = \vec{V}t + \vec{X}_0 \quad (X_0 = \text{const. of integration}).
\end{aligned}$$

(c)

$$\begin{aligned}
\vec{J} &= \sum_j \vec{x}_j \wedge \vec{p}_j \\
\Rightarrow \dot{\vec{J}} &= \sum_j \dot{\vec{x}}_j \wedge \vec{p}_j + \sum_j \vec{x}_j \wedge \dot{\vec{p}}_j
\end{aligned} \tag{4.2}$$

Now $\vec{p}_j = m_j \dot{\vec{x}}_j \Rightarrow \dot{\vec{x}}_j \wedge \vec{p}_j = 0 \quad \forall j$. Hence (4.2) becomes:

$$\begin{aligned}
\dot{\vec{J}} &= \sum_j \vec{x}_j \wedge \dot{\vec{p}}_j \\
&= \sum_j \vec{x}_j \wedge \vec{k}_j \\
&= \sum_j \vec{x}_j \wedge \left\{ \sum_{i \neq j} \vec{k}_{ji} + \vec{K}_j \right\}.
\end{aligned} \tag{4.3}$$

Now

$$\sum_j \vec{x}_j \wedge \sum_{i \neq j} \vec{k}_{ji} = 0$$

since, for any $i \neq j$ (using centrality here), force is along line joining the bodies:

$$\begin{aligned}
\vec{x}_j \wedge \vec{k}_{ji} &= \vec{x}_j \wedge \frac{\vec{x}_j - \vec{x}_i}{|\vec{x}_j - \vec{x}_i|} f_{ji}(|\vec{x}_j - \vec{x}_i|), \\
\vec{x}_i \wedge \vec{k}_{ij} &= \vec{x}_i \wedge \frac{\vec{x}_i - \vec{x}_j}{|\vec{x}_i - \vec{x}_j|} f_{ij}(|\vec{x}_i - \vec{x}_j|).
\end{aligned}$$

Their sum is

$$\begin{aligned}
(\vec{x}_j \wedge \vec{k}_{ji}) + (\vec{x}_i \wedge \vec{k}_{ij}) &= (\vec{x}_j - \vec{x}_i) \wedge \frac{\vec{x}_j - \vec{x}_i}{|\vec{x}_j - \vec{x}_i|} f_{ij}(|\vec{x}_j - \vec{x}_i|) \\
&= 0 \quad \text{as } f_{ij} = f_{ji} \quad (\text{Newton's 3rd}).
\end{aligned}$$

Thus (4.3) becomes:

$$\dot{\vec{J}} = \sum_j \vec{x}_j \wedge \vec{K}_j.$$

(d)
(i)

$$\bar{k}_1 = m_1 \ddot{\bar{x}}_1 = \frac{\bar{x}_1 - \bar{x}_2}{|\bar{x}_1 - \bar{x}_2|} f_{12}(|\bar{x}_1 - \bar{x}_2|)$$

$$\bar{k}_2 = m_2 \ddot{\bar{x}}_2 = \frac{\bar{x}_2 - \bar{x}_1}{|\bar{x}_1 - \bar{x}_2|} f_{21}(|\bar{x}_2 - \bar{x}_1|).$$

$$\Rightarrow \quad \ddot{\bar{x}}_1 = \frac{1}{m_1} \frac{\bar{x}_1 - \bar{x}_2}{|\bar{x}_1 - \bar{x}_2|} f_{12}(|\bar{x}_1 - \bar{x}_2|)$$

$$\ddot{\bar{x}}_2 = \frac{1}{m_2} \frac{\bar{x}_2 - \bar{x}_1}{|\bar{x}_2 - \bar{x}_1|} f_{21}(|\bar{x}_2 - \bar{x}_1|).$$

$$\Rightarrow \quad \ddot{\bar{x}}_2 - \ddot{\bar{x}}_1 = \frac{1}{m_2} \frac{\bar{x}_2 - \bar{x}_1}{|\bar{x}_2 - \bar{x}_1|} f_{21}(|\bar{x}_2 - \bar{x}_1|) + \frac{1}{m_1} \frac{\bar{x}_2 - \bar{x}_1}{|\bar{x}_2 - \bar{x}_1|} f_{12}(|\bar{x}_2 - \bar{x}_1|)$$

$$= \frac{m_1 + m_2}{m_1 m_2} \frac{\bar{x}_2 - \bar{x}_1}{|\bar{x}_2 - \bar{x}_1|} f_{12}(|\bar{x}_2 - \bar{x}_1|).$$

$$\Rightarrow \quad \frac{m_1 m_2}{m_1 + m_2} (\ddot{\bar{x}}_2 - \ddot{\bar{x}}_1) = \frac{\bar{x}_2 - \bar{x}_1}{|\bar{x}_2 - \bar{x}_1|} f_{12}(|\bar{x}_2 - \bar{x}_1|)$$

i.e.

$$\mu \ddot{\bar{r}} = \frac{\bar{r}}{|\bar{r}|} f_{12}(|\bar{r}|).$$

It should be noted that $f_{12}(|r|) = f_{21}(|r|)$ must be the case for Newton's 3rd law to hold.

(ii)

$$T = \frac{1}{2} m_1 \dot{\bar{x}}_1^2 + \frac{1}{2} m_2 \dot{\bar{x}}_2^2$$

Now

$$\bar{X} = \frac{1}{m_1 + m_2} [m_1 \bar{x}_1 + m_2 \bar{x}_2]$$

$$\Rightarrow \quad \dot{\bar{X}} = \frac{1}{m_1 + m_2} [m_1 \dot{\bar{x}}_1 + m_2 \dot{\bar{x}}_2]$$

$$\Rightarrow \quad |\dot{\bar{X}}|^2 = \dot{\bar{X}}^2 = \frac{1}{(m_1 + m_2)^2} [m_1^2 \dot{\bar{x}}_1^2 + 2m_1 m_2 \dot{\bar{x}}_1 \cdot \dot{\bar{x}}_2 + m_2^2 \dot{\bar{x}}_2^2]$$

$$\Rightarrow \quad \frac{1}{2} (m_1 + m_2) \dot{\bar{X}}^2 = \frac{1}{2(m_1 + m_2)} [m_1^2 \dot{\bar{x}}_1^2 + 2m_1 m_2 \dot{\bar{x}}_1 \cdot \dot{\bar{x}}_2 + m_2^2 \dot{\bar{x}}_2^2]$$

Hence, using $M = (m_1 + m_2)$, we obtain:

$$\begin{aligned}
T - \frac{1}{2}M\dot{X}^2 &= \left[\frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 \right] - \frac{m_1^2}{2(m_1 + m_2)}\dot{x}_1^2 - \frac{m_1m_2}{m_1 + m_2}\dot{x}_1 \cdot \dot{x}_2 - \frac{m_2^2}{2(m_1 + m_2)}\dot{x}_2^2 \\
&= \frac{1}{2}\dot{x}_1^2 \left[m_1 - \frac{m_1^2}{m_1 + m_2} \right] + \frac{1}{2}\dot{x}_2^2 \left[m_2 - \frac{m_2^2}{m_1 + m_2} \right] - \frac{m_1m_2}{m_1 + m_2}\dot{x}_1 \cdot \dot{x}_2 \\
&= \frac{1}{2} \frac{m_1m_2}{m_1 + m_2} [\dot{x}_1^2 - 2\dot{x}_1 \cdot \dot{x}_2 + \dot{x}_2^2] \\
&= \frac{1}{2} \frac{m_1m_2}{m_1 + m_2} (\dot{x}_2 - \dot{x}_1)^2 \\
&= \frac{1}{2}\mu\dot{r}^2
\end{aligned}$$

i.e

$$T = \frac{1}{2}M\dot{X}^2 + \frac{1}{2}\mu\dot{r}^2$$

(iii)

$$\bar{J} = \bar{x}_1 \wedge m_1\dot{x}_1 + \bar{x}_2 \wedge m_2\dot{x}_2$$

Now

$$\begin{aligned}
M\bar{X} \wedge \dot{X} &= (m_1 + m_2) \frac{1}{m_1 + m_2} (m_1\bar{x}_1 + m_2\bar{x}_2) \wedge \frac{1}{m_1 + m_2} (m_1\dot{x}_1 + m_2\dot{x}_2) \\
&= \frac{1}{m_1 + m_2} (m_1\bar{x}_1 + m_2\bar{x}_2) \wedge (m_1\dot{x}_1 + m_2\dot{x}_2) \\
&= \frac{1}{m_1 + m_2} [m_1^2\bar{x}_1 \wedge \dot{x}_1 + m_2^2\bar{x}_2 \wedge \dot{x}_2 + m_1m_2\bar{x}_1 \wedge \dot{x}_2 + m_1m_2\bar{x}_2 \wedge \dot{x}_1]
\end{aligned}$$

Hence:

$$\begin{aligned}
\bar{J} - M\bar{X} \wedge \dot{X} &= [m_1\bar{x}_1 \wedge \dot{x}_1 + m_2\bar{x}_2 \wedge \dot{x}_2] \\
&\quad - \frac{1}{m_1 + m_2} [m_1^2\bar{x}_1 \wedge \dot{x}_1 + m_2^2\bar{x}_2 \wedge \dot{x}_2 + m_1m_2\bar{x}_1 \wedge \dot{x}_2 + m_1m_2\bar{x}_2 \wedge \dot{x}_1] \\
&= \frac{m_1m_2}{m_1 + m_2} [\bar{x}_1 \wedge \dot{x}_1 + \bar{x}_2 \wedge \dot{x}_2 - \bar{x}_1 \wedge \dot{x}_2 - \bar{x}_2 \wedge \dot{x}_1] \\
&= \frac{m_1m_2}{m_1 + m_2} (\bar{x}_2 - \bar{x}_1) \wedge (\dot{x}_2 - \dot{x}_1) \\
&= \mu\bar{r} \wedge \dot{r}
\end{aligned}$$

Hence:

$$\bar{J} = M\bar{X} \wedge \dot{X} + \mu\bar{r} \wedge \dot{r}$$